Blessing Chikumbirike

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PROFESSIONAL SUMMARY

Data Science and Informatics graduate (BSc Honours, Upper Second Class) who builds machine learning systems from raw data to deployed web application. Delivered an explainable deep learning model for chest X-ray classification (89.74% accuracy; AUC 0.9589) inside a full-stack Flask application. Strong in Python, SQL, Excel and data visualization; comfortable writing technical documentation for both academic and non-technical audiences. Seeking remote junior data science, machine learning or computer vision roles.

PROJECTS AND EXPERIENCE

PneumoInsight: Explainable Pneumonia Detection System 2025 – 2026

University of Zimbabwe, BSc Honours Capstone

- Trained and benchmarked three architectures (DenseNet121, ResNet-50 and a custom CNN) on chest X-ray images; selected DenseNet121 as the production model with 89.74% accuracy and an AUC of 0.9589.
- Implemented Guided Grad-CAM so each prediction is paired with a visual explanation clinicians can inspect.
- Built a full-stack Flask web application with role-based access control, automated PDF report generation and audit logging.
- Authored a six-chapter thesis and an IEEE-style technical paper; presented the work in a 17-slide oral defense.

Data Analyst Ongoing

Independent Projects

- Built web scraping pipelines with BeautifulSoup, improving data acquisition efficiency by 30%.
- Performed exploratory data analysis in Python and Pandas to surface insights that support decision-making.
- Developed interactive dashboards in Power BI and Tableau to improve stakeholder engagement with data.
- Optimized MySQL queries, improving retrieval speed by 25% on large datasets.
- Portfolio: SQL data cleaning and exploratory analysis of a global layoffs dataset (MySQL); interactive Tableau dashboard analyzing Airbnb market trends.

EDUCATION

BSc (Honours) Data Science and Informatics, Upper Second Class (2.1) 2026

University of Zimbabwe, Department of Analytics and Informatics

- Coursework and applied work in machine learning, deep learning, computer vision and data management.

TECHNICAL SKILLS

Programming and Web	Python, Flask, SQLAlchemy, Flask-Migrate, REST APIs
Machine Learning	CNNs, DenseNet121, ResNet-50, custom CNN architectures, transfer learning, TensorFlow/Keras, Guided Grad-CAM, model evaluation and explainability
Data Analysis	Excel (advanced), Pandas, NumPy, exploratory data analysis, data cleaning and preparation, web scraping (BeautifulSoup)
Data and Databases	PostgreSQL, MySQL, SQL, query optimization, Alembic
Visualization and BI	Power BI, Tableau, Excel dashboards
Cloud and Tools	AWS, Git/GitHub, VS Code